# CLOUD COMPUTING
# Cloud Service Models & IAM

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**Topic:** IAM & Cloud Service Models
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# Cloud Service Models & IAM

## Introduction to Cloud Computing

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, analytics, and intelligence over the internet ("the cloud"). Instead of owning and maintaining physical hardware or data centers, organizations can access these resources on demand from cloud providers. This allows businesses and developers to scale resources easily, reduce infrastructure costs, and access services from anywhere with an internet connection.

Major cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform offer different service models that allow users to choose the level of control and management they need.

## Part 1: Understanding the Cloud service Models

The three main cloud service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). Each model provides a different level of responsibility between the cloud provider and the user.

### 1. Infrastructure as a Service (IaaS)

Infrastructure as a Service provides fundamental computing resources such as virtual machines, storage, and networking over the internet. The cloud provider manages the physical infrastructure, while the user is responsible for managing the operating system, applications, and data.

AWS Example:

Amazon EC2 (Elastic Compute Cloud) allows users to launch virtual servers in the cloud and run applications just like a physical server.

## 2. Platform as a Service (PaaS)

Platform as a Service provides a development platform where developers can build, test, and deploy applications without worrying about managing servers, operating systems, or infrastructure.

AWS Example:

AWS Elastic Beanstalk automatically handles deployment, scaling, load balancing, and application monitoring.

## 3. Software as a Service (SaaS)

Software as a Service provides ready-to-use applications that are delivered over the internet. Users can access these applications using a web browser without installing or maintaining any software.

AWS Example:

Amazon WorkDocs is a cloud-based file storage and collaboration service that allows teams to share and manage documents online.

## Difference Between Cloud Service Models

| Feature | IaaS | PaaS | SaaS |
|---|---|---|---|
| Control Level | High | Medium | Low |
| Managed by Cloud Provider | Hardware | Hardware + OS + Runtime | Everything |
| Managed by User | OS, Applications, Data | Applications and Data | Only Usage |
| Example | Amazon EC2 | AWS Elastic Beanstalk | Amazon WorkDocs |

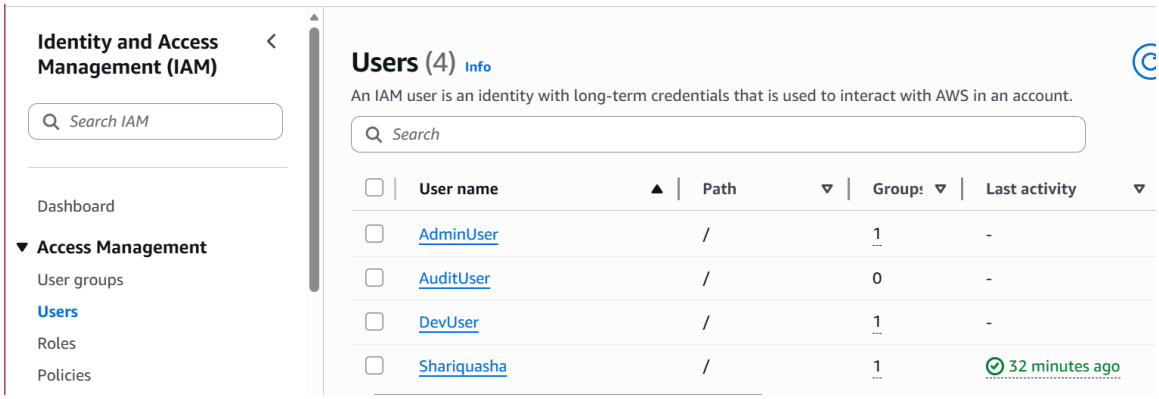
## Part 2: IAM Hands-On Activity

AWS Identity and Access Management (IAM) is a service that helps administrators securely control access to AWS resources. Using IAM, organizations can create users, groups, and policies to manage who can access specific AWS services.

### Step 1: Create IAM Users

The following IAM users were created:

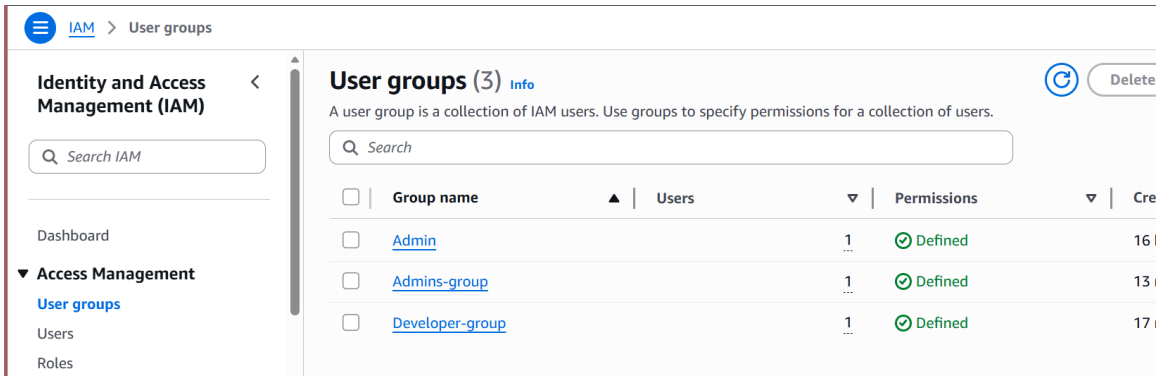
- DevUser
- AdminUser
- AuditUser



## Step 2: Create IAM Groups

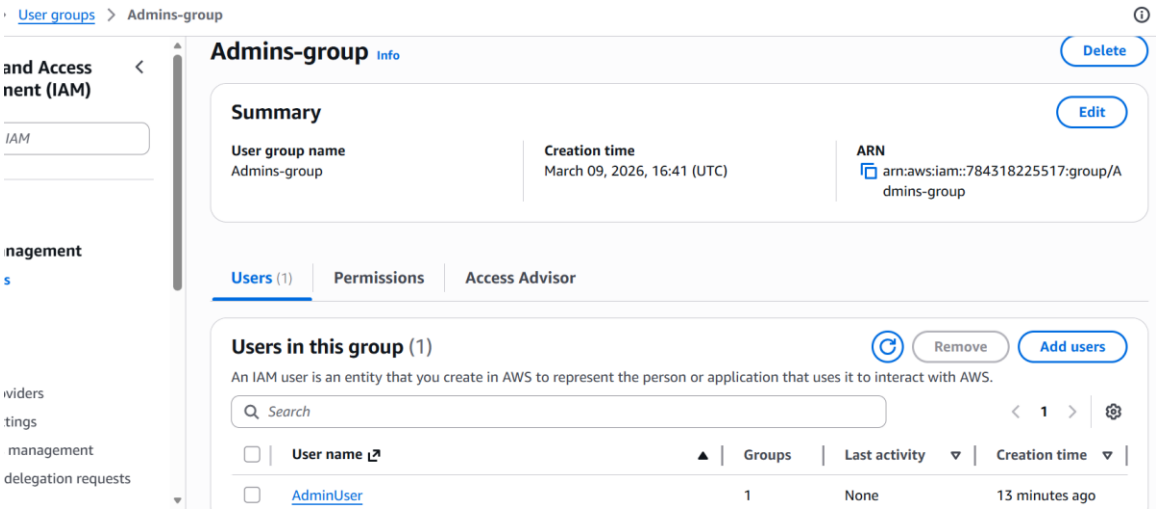
Two IAM groups were created:

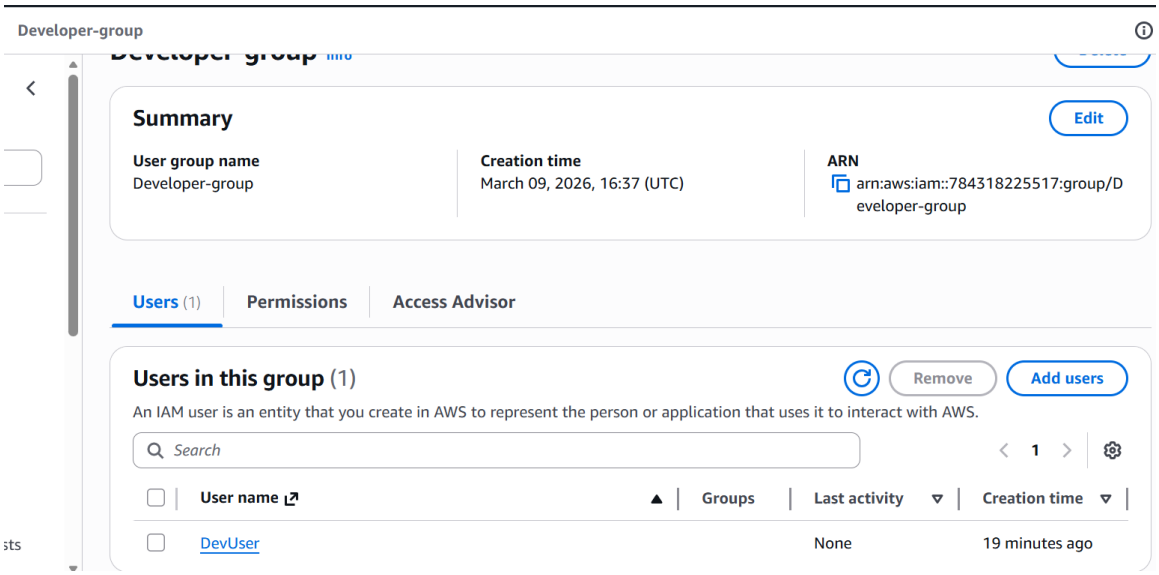
- Developers
- Admins




## Step 3: Assign Users to Groups

DevUser was assigned to the **Developers** group, while AdminUser was assigned to the **Admins** group to simplify permission management.

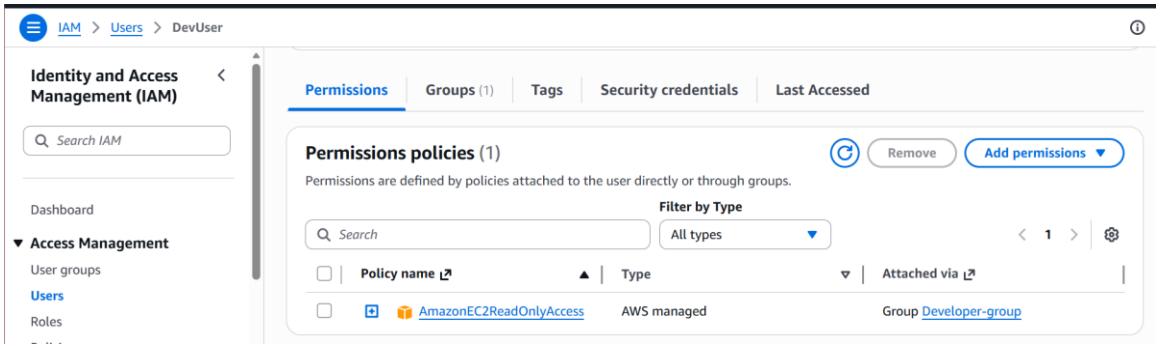




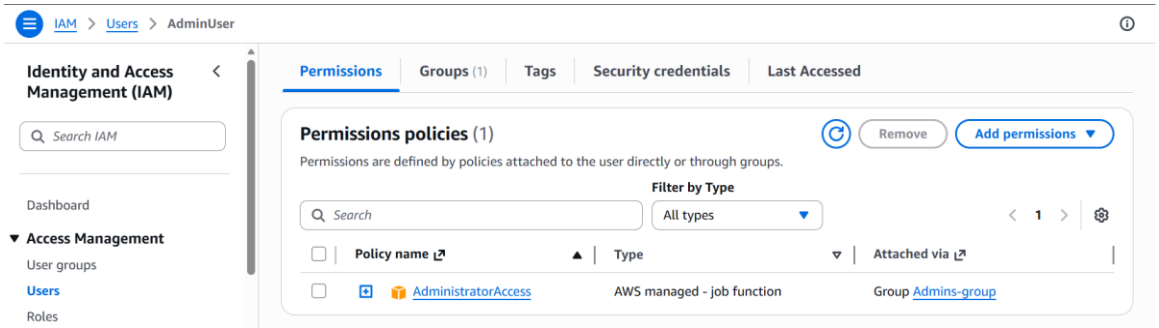

## Step 4: Attach IAM Policies

Policies were attached to define access permissions.

DevUser was assigned the AmazonEC2ReadOnlyAccess policy which allows viewing EC2 resources without modifying them.

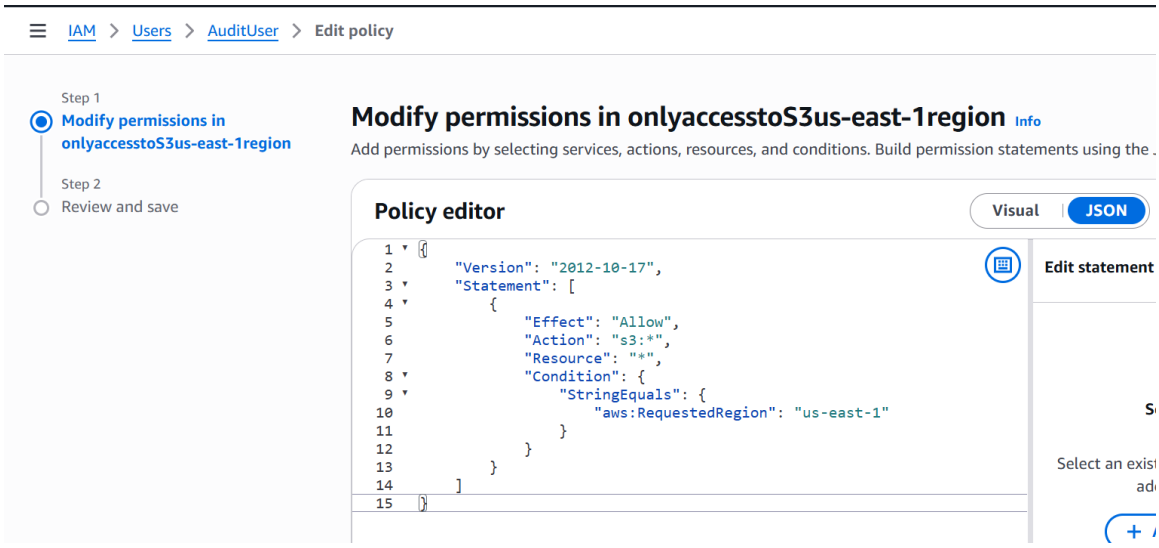


AdminUser was assigned the AdministratorAccess policy which provides full administrative access to AWS services.

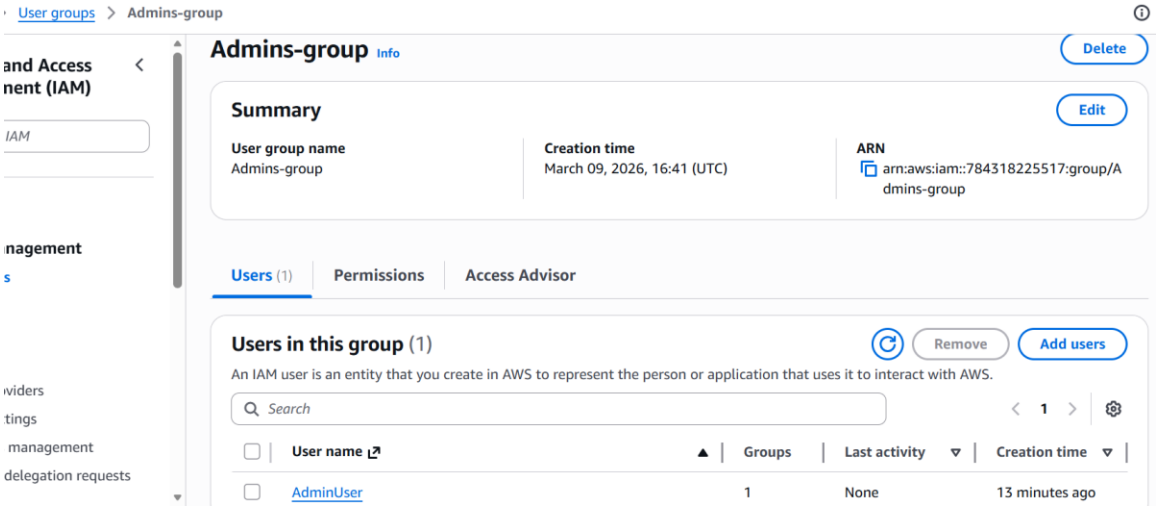


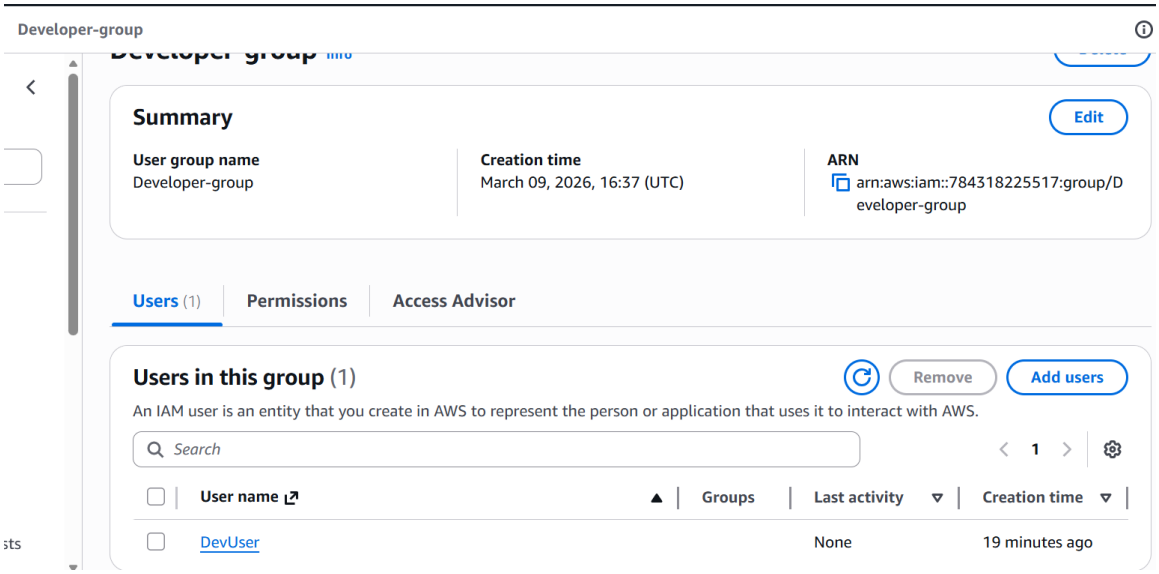

## AuditUser Inline Policy

For AuditUser, an inline policy was created to allow access only to Amazon S3 resources in the us-east-1 region. This restriction ensures that the user can audit storage resources only within the specified region.

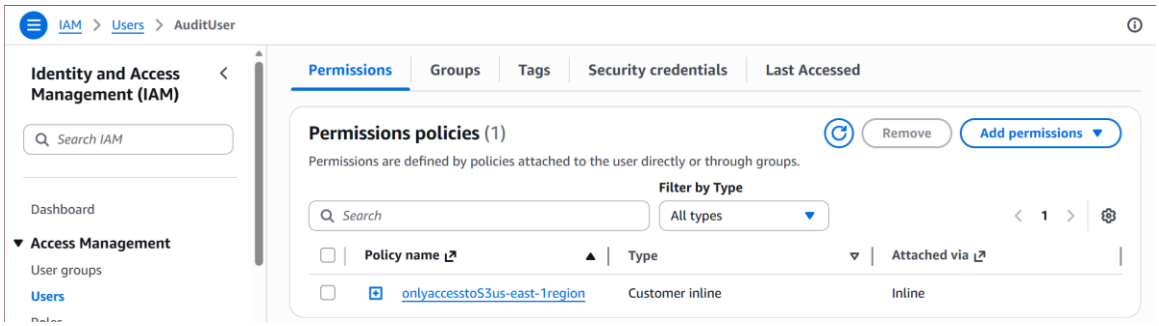


## Screenshots




IAM Console showing created users and their assigned groups




Inline policy attached to AuditUser

## Conclusion

In conclusion, cloud computing provides flexible and scalable computing resources through different service models such as IaaS, PaaS, and SaaS. Each model offers a different level of control and responsibility between the user and the cloud provider. For example, services like Amazon EC2 provide infrastructure resources, while platform services such as AWS Elastic Beanstalk simplify application deployment, and software solutions like Amazon WorkDocs deliver ready-to-use applications over the internet.

Additionally, the hands-on activity with AWS Identity and Access Management demonstrated how organizations manage user permissions securely in a cloud environment. By creating IAM users, groups, and policies, administrators can control access to AWS resources effectively. The use of an inline policy for Amazon S3 also showed how access can be restricted to a specific region to improve security and governance. This exercise helped in understanding both the theoretical concepts of cloud service models and the practical implementation of access management in AWS.